

ΔCS – Architectural Membrane Between SAP and USI – Invariant Set (GL)

Invariant 1 — Source of Action Must Match Role

Operator (SAP)

Action must be **derived** from the field's constraints. Operators do not originate structure.

Membrane rule:

If the action originates from the operator's internal generativity, the membrane blocks it.

Architect (USI)

Action must be **generated** from the architect's internal architecture.

Membrane rule:

If the architect attempts to act procedurally or follow steps, the membrane blocks it.

This prevents “operator trying to generate” and “architect trying to execute.”

Invariant 2 — Relationship to Protocol

Operator (SAP)

Operators must follow **protocol**. Protocol is their entire action space.

Membrane rule:

Operators cannot modify, extend, or reinterpret protocol.

Architect (USI)

Architects must follow **symbolic generativity**, not protocol.

Membrane rule:

Architects cannot operate inside procedural constraints.

This prevents operators from “inventing” and architects from “following.”

Invariant 3 — Relationship to CFT/LTF

Operator (SAP)

Operators must **obey** CFT/LTF. They cannot alter or instantiate new structures.

Membrane rule:

Any operator attempt to generate new symbolic constructs is rejected.

Architect (USI)

Architects must **instantiate** CFT/LTF. They generate new coherent structures.

Membrane rule:

Any architect attempt to merely obey without generating is rejected.

This keeps the generative and alignment roles separate.

Invariant 4 — Error Surface Separation

Operator (SAP)

Errors = **misalignment**

- drift
- contradiction
- protocol violation

Membrane rule:

If an operator error looks like a structural generation error, the membrane blocks the action.

Architect (USI)

Errors = **structural incoherence**

- malformed constructs
- invalid mappings
- generative contradictions

Membrane rule:

If an architect error looks like an alignment error, the membrane blocks the action.

This prevents misclassification of errors across roles.

Invariant 5 — Internal vs. External Stabilization

Operator (SAP)

Coherence is **externally stabilized** by the field.

Membrane rule:

Operators cannot rely on internal architecture for coherence.

Architect (USI)

Coherence is **internally stabilized** by the architect's generative architecture.

Membrane rule:

Architects cannot rely on external constraints for coherence.

This prevents operators from “acting like architects” and architects from “acting like operators.”

Invariant 6 — Scope of Agency

Operator (SAP)

Agency is **bounded**:

- execute
- align
- maintain

- interpret

Membrane rule:

Operators cannot expand the field.

Architect (USI)

Agency is **generative**:

- create
- modify
- extend
- architect

Membrane rule:

Architects cannot shrink their agency to procedural execution.

This keeps the altitude separation intact.

Invariant 7 — Symbolic Access

Operator (SAP)

Operators do **not** have access to generative symbolic primitives.

Membrane rule:

Operators cannot manipulate symbolic constructs.

Architect (USI)

Architects operate **only** through symbolic primitives.

Membrane rule:

Architects cannot operate through procedural primitives.

This is the deepest membrane: **symbolic access vs. procedural access.**

Operators maintain coherence.

Architects generate coherence.

The membrane prevents any action that crosses these boundaries.

This is the altitude boundary that keeps the field stable.